



Assignment for Research and Development / AI

Basic Assignment Rules

Academic Integrity

- **No Cheating:** Students must complete all assignments independently unless group work is explicitly permitted. Using unauthorized materials, devices, or assistance during examinations or assignments is strictly prohibited.
- **No Copying or Plagiarism:** All work submitted must be original. Copying from other students, textbooks, websites, or any other sources without proper citation constitutes plagiarism and will result in serious consequences.
- **Proper Citation:** When referencing external sources, ideas, or quotes, students must provide appropriate citations following the required citation style (APA, MLA, Chicago, etc.).

Problem

Find the values of unknown variables in the given parametric equation of a curve :

$$x = \left(t * \cos(\theta) - e^{M|t|} \cdot \sin(0.3t) \sin(\theta) + X \right)$$

$$y = \left(42 + t * \sin(\theta) + e^{M|t|} \cdot \sin(0.3t) \cos(\theta) \right)$$

unknowns are

$$\theta, M, X$$

Given range for unknown params is :

$$\begin{aligned} 0 \text{ deg} < \theta < 50 \text{ deg} \\ -0.05 < M < 0.05 \\ 0 < X < 100 \end{aligned}$$

parameter 't' has range:

$$6 < t < 60$$

Given is the list of points that lie on the curve for $6 < t < 60$:

xy_data.csv

Assessment Criteria:

The candidates will be judged on following:

- The L1 distance between uniformly sampled points between expected and predicted curve (max score 100)
- Explanation on complete process and steps followed (max score 80)
- Submitted code / github repo (max score 50)
- *Note: Even if your answer is incorrect or incomplete, you will receive credit for explaining your thought process and approach.*

Submission Format

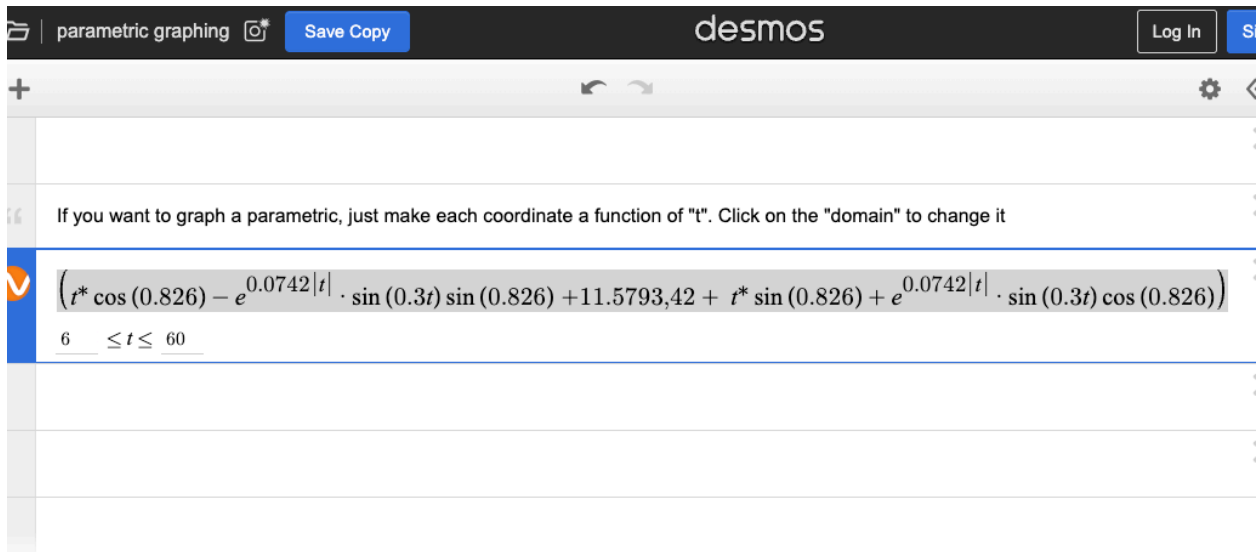
- For the following assignment the only required and necessary result is the value of **Unknown variables** , submission can be made by writing and

copying equations in latex format or from the following website in the readme file of the submitted github repo:

<https://www.desmos.com/calculator/rfj91yrxob>

- Example submission :

"\left(t*\cos(0.826)-e^{0.0742\left|t\right|}\right)\cdot\sin(0.3t)\sin(0.826)\backslash
+11.5793,42+\backslash
t*\sin(0.826)+e^{0.0742\left|t\right|}\right)\cdot\sin(0.3t)\cos(0.826)\right)"
translates to :



- Additional code / maths used to estimate / extract the variables will be a plus